KDM5A fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes KDM5A gene fusions from the msk_impact_50k_2026 study, covering 10 fusion events. 3/6 fusions with known frame status (50.0%) were in-frame. 5/9 fusions (55.6%) retained the PLU-1-like protein domain. Domain retention was tested with Fisher's exact test (p=0.119048, not statistically significant at alpha=0.05). No literature benchmark is configured for KDM5A.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 10 analyzed fusion events, identifying 9 distinct fusion partner genes. The most frequent partner was NINJ2, observed in 2 events.

Domain retention

PLU-1-like protein domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/3 (100.0%) of in-frame fusions retained the domain, p=0.119048 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.04995.

Domain disruption

Disruption of the jmjN domain, ARID/BRIGHT DNA binding domain, PHD-finger, JmjC domain, hydroxylase, and C5HC2 zinc finger was tested with Fisher's exact test comparing in-frame fusions against all others; 1/3 (33.3%) of in-frame fusions disrupted the domain, p=0.761905 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 1.

Junction concentration

33% of 3 fusions with this gene as the 3' partner join at exon 24, against 0% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.217822, corrected for testing every exon, not statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 9 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1311 aa (exon 24; label-permutation p=0.215784). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is 94 aa from the nearest configured domain boundary at 1217 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=5) and "not_retained" (n=4). For Frame_status, retained: 3/4 (75.0%, 95% CI 30.1%-95.4%); not_retained: 0/2 (0.0%, 95% CI 0.0%-65.8%). Welch's t-test comparing Tumor_variant_count between groups gave means of 15.6 vs 17.25, p=0.843016 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for KDM5A in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (8 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 0 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: KDM5A has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: KDM5A has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
NINJ2	2	0.2	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.14937302161590618	1
BMAL2	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	2
CACNA1C	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	3
COLGALT1	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	4
DHX37	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	5
DIPK1A	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	6
OSBPL8	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	7
STAT5A	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	8
USHBP1	1	0.1	0.13014640731432997	0.0	None	0.34430466544937777	confidence_certainty, domain_disruption, domain_retention, recurrence	0.11407890396884737	9

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
189	0	5	1	3	0.0	0.4444444444444445	0.3521825181113625
224	0	5	2	2	0.0	0.16666666666666669	0.7781512503836436
225	1	4	2	2	0.25	0.523809523809524	0.2808266095756941
664	2	3	2	2	0.6666666666666666	1.0	-0.0
759	2	3	3	1	0.2222222222222222	0.5238095238095237	0.2808266095756943
1311	2	3	4	0	0.0	0.16666666666666663	0.7781512503836437
1358	3	2	4	0	0.0	0.4444444444444444	0.35218251811136253
1485	4	1	4	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

1	2
2	4

frame_domain_contingency_tables_by_tumor_type

stratum	table
CCOV	[0, 0], [0, 1]
GEJ	[0, 0], [1, 0]
HGSOC	[0, 1], [0, 0]
IDC	[1, 0], [0, 0]
LMS	[0, 0], [1, 0]
LUAD	[0, 1], [0, 1]
MFH	[0, 0], [0, 1]
PRAD	[0, 0], [0, 1]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf02375	jmjN domain	9	6	0	3	0.0
auto_disruption_pf01388	ARID/BRIGHT DNA binding domain	9	6	0	3	0.0
auto_disruption_pf00628	PHD-finger	9	5	0	4	0.0
auto_disruption_pf02373	JmjC domain, hydroxylase	9	5	0	4	0.0
auto_disruption_pf02928	C5HC2 zinc finger	9	6	0	3	0.0

domain_retention tables

frame_domain_contingency_table

3	2
0	4

frame_domain_contingency_tables_by_tumor_type

stratum	table
CCOV	[0, 0], [0, 1]
GEJ	[1, 0], [0, 0]
HGSOC	[0, 0], [0, 1]
IDC	[1, 0], [0, 0]
LMS	[1, 0], [0, 0]
LUAD	[0, 1], [0, 1]
MFH	[0, 1], [0, 0]
PRAD	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	PLU-1-like protein	9	5	1	3	0.015015015015015015

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
16	6	10	0.6

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
BMAL2	1	1	1	1
CACNA1C	1	1	1	1
COLGALT1	1	1	1	1
DHX37	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
DIPK1A	1	1	1	1
NINJ2	2	2	2	2
OSBPL8	1	1	1	1
STAT5A	1	1	1	1
USHBP1	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
12	432000	433000	2	EVT-P-0030430-T03-IM7-42, EVT-P-0030719-T01-IM6-4
12	401000	402000	1	EVT-P-0060766-T02-IM7-41
12	402000	403000	1	EVT-P-0064343-T01-IM7-8
12	415000	416000	1	EVT-P-0065027-T03-IM7-9
12	416000	417000	1	EVT-P-0021409-T01-IM6-1
12	465000	466000	1	EVT-P-0030473-T01-IM6-5
12	470000	471000	1	EVT-P-0039045-T01-IM6-44
12	472000	473000	1	EVT-P-0022798-T01-IM6-43

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	5	2	6	0.3333333333333333	0.06000402466019062	1.3378995164098522
five_prime	16	1	6	0.16666666666666666	0.005844904231381373	1.4614142287785212
five_prime	24	1	6	0.16666666666666666	0.09045423778424162	0.3626810389637204
five_prime	26	1	6	0.16666666666666666	0.024001609864076248	0.8676118817131107
five_prime	27	1	6	0.16666666666666666	0.06866619102500778	0.4591422792975807
three_prime	6	1	3	0.3333333333333333	0.059738762965808684	0.7728248864577678
three_prime	15	1	3	0.3333333333333333	0.04788430931342956	0.8636473093060405
three_prime	24	1	3	0.3333333333333333	0.0046466531291732985	1.8577580701478102

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021409-T01-IM6-1	P-0021409-T01-IM6	P-0021409	Ovarian Cancer	HGSOC	COLGALT1-KDM5A	COLGALT1	unknown	three_prime	416253	24	1311	False	1311	True	exact match	lost	0.0	False		PLU-1-like protein; jmjN domain; ARID/BRIGHT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger		COLGALT1 (NM_024656)-KDM5A (NM_001042603) rearrangement: t(19;12)(p13.11;p13.33)(chr19:g.17669210::chr12:g.416253) | Protein Fusion: mid-exon {COLGALT1:KDM5A}	NA
EVT-P-0030719-T01-IM6-4	P-0030719-T01-IM6	P-0030719	Non-Small Cell Lung Cancer	LUAD	KDM5A-BMAL2	BMAL2	unknown	three_prime	432924	15	664	False	664	True	exact match	retained	1.0	False	PLU-1-like protein; PHD-finger; C5HC2 zinc finger	jmjN domain; ARID/BRIGHT DNA binding domain; JmjC domain, hydroxylase		ARNTL2 (NM_001248002) - KDM5A (NM_001042603) rearrangement: c.285-4357:ARNTL2_c.1992:KDM5A | Protein Fusion: mid-exon {ARNTL2:KDM5A}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030473-T01-IM6-5	P-0030473-T01-IM6	P-0030473	Breast Cancer	IDC	KDM5A-CACNA1C	CACNA1C	in-frame	three_prime	465876	6	225	True	225	True	exact match	retained	1.0	False	PLU-1-like protein; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger	jmjN domain; ARID/BRIGHT DNA binding domain		CACNA1C (NM_0011298 27) - KDM5A (NM_001042603) rearrangement: c.1114-2453:CACNA1C_c.673-173:KDM5Ainv | Protein Fusion: in frame {CACNA1C:KDM5A}	NA
EVT-P-0064343-T01-IM7-8	P-0064343-T01-IM7	P-0064343	Soft Tissue Sarcoma	MFH	KDM5A-DHX37	DHX37	out-of-frame	five_prime	402339	26	1485	True	1485	True	exact match	retained	1.0	False	PLU-1-like protein; jmjN domain; ARID/BRIGHT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger			KDM5A (NM_001042603) - DHX37 (NM_032656) rearrangement: c.4456-4:KDM5A_c.2045+567:DHX37dup | Protein Fusion: out of frame {KDM5A:DHX37}	NA

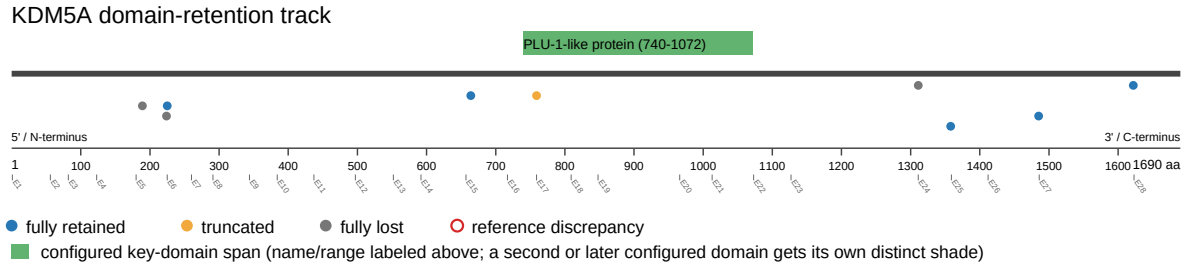
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065027-T03-IM7-9	P-0065027-T03-IM7	P-0065027	Esophagogastric Cancer	GEJ	KDM5A-DIPK1A	DIPK1A	in-frame	five_prime	415999	24	1358	True	1358	True	exact match	retained	1.0	False	PLU-1-like protein; jmjN domain; ARID/BRIGHT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger			KDM5A (NM_001042603) - FAM69A (NM_001006605) rearrangement: t(1;12)(p22.1;p13.33)(chr1:g.93411462::chr12:g.415999) | Protein Fusion: in frame {KDM5A:FAM69A}	NA
EVT-P-0060766-T02-IM7-41	P-0060766-T02-IM7	P-0060766	Soft Tissue Sarcoma	LMS	KDM5A-NINJ2	NINJ2	in-frame	five_prime	401803	27	1622	True	1622	True	exact match	retained	1.0	False	PLU-1-like protein; jmjN domain; ARID/BRIGHT DNA binding domain; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger			KDM5A (NM_001042603) - NINJ2 (NM_016533) rearrangement: c.4866+122:KDM5A_c.172-28503:NINJ2dup | Protein Fusion: in frame {KDM5A:NINJ2}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030430-T03-IM7-42	P-0030430-T03-IM7	P-0030430	Non-Small Cell Lung Cancer	LUAD	KDM5A-NINJ2	NINJ2	out-of-frame	five_prime	432137	16	759	True	758	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.0600600600606	True	jmjN domain; ARID/BRIGHT DNA binding domain; JmjC domain, hydroxylase; C5HC2 zinc finger	PHD-finger	PLU-1-like protein	KDM5A (NM_001042603) - NINJ2 (NM_016533) rearrangement: c.2275+111:KDM5A_c.172-20595:NINJ2dup | Protein Fusion: out of frame {KDM5A:NINJ2}	NA
EVT-P-0022798-T01-IM6-43	P-0022798-T01-IM6	P-0022798	Prostate Cancer	PRAD	KDM5A-OSBPL8	OSBPL8	unknown	five_prime	472234	5	189	False	189	True	exact match	lost	0.0	False	jmjN domain; ARID/BRIGHT DNA binding domain	PLU-1-like protein; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger		KDM5A (NM_001042603) - OSBPL8 (NM_020841) Rearrangement : c.567:KDM5A_c.1630+2182:OSBPL8dup | Protein Fusion: mid-exon {KDM5A:OSBPL8}	NA

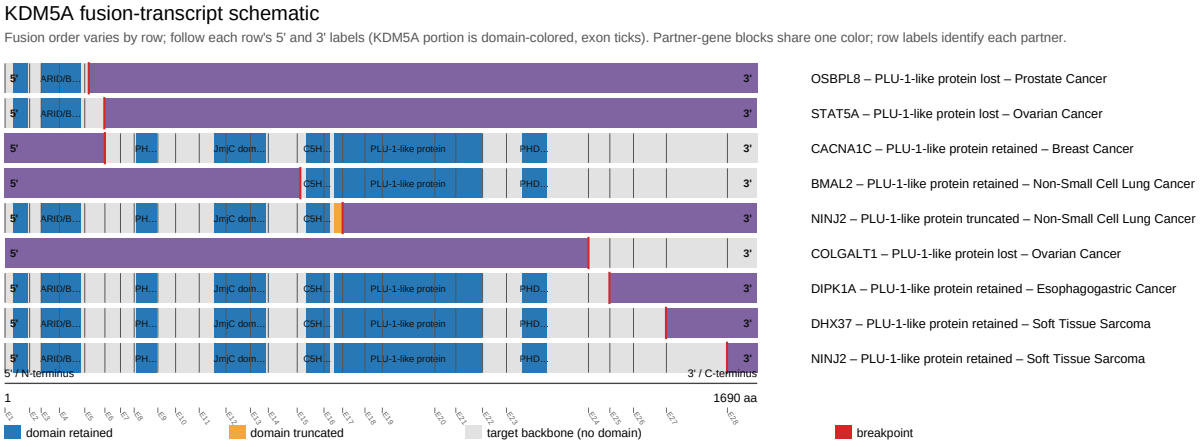
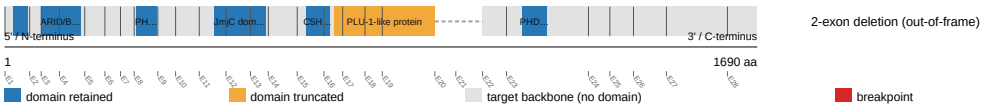
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039045-T01-IM6-44	P-0039045-T01-IM6	P-0039045	Ovarian Cancer	CCOV	KDM5A-STAT5A	STAT5A	out-of-frame	five_prime	470840	5	224	True	224	True	exact match	lost	0.0	False	jmjN domain; ARID/BRIGHT DNA binding domain	PLU-1-like protein; PHD-finger; JmjC domain, hydroxylase; C5HC2 zinc finger		KDM5A (NM_001042603) - STAT5A (NM_003152) rearrangement: t(12;17)(p13.33;q21.2)(chr12:g.470840::chr17:g.40459887) | Protein Fusion: out of frame {KDM5A:STAT5A}	NA

Figures

domain retention outliers



fusion schematic



intragenic deletion schematic